

Reference

Electronic device } Churba
Electronic circuits } Banjade

Principle of Electronics
- V. K. mehta

Numerical + Theory

Total chapter - 9

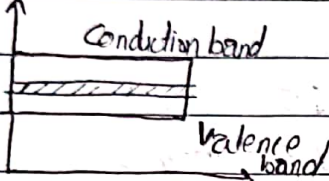
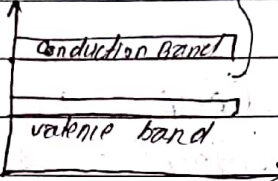
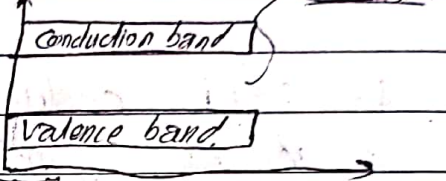
1. Semi conductor diode (6) - 15 marks
2. Bipolar junction transistor (6) - 15 marks
3. D.C power supply (5) - 12/13 marks
4. field Effect transistor (5) - 12/13 marks
5. Small signal low frequency analysis model of BJT (5)
↳ 12/13 marks
6. Multistage Amplifier (4) - 7/8 marks
7. large signal amplifier (4) - 7/8 marks
8. feedback Amplifier (3) - 7/8 marks
9. Operational Amplifier and oscillator (7) - 15 marks

(1 Numerical + 1 Theory) $\Rightarrow$ 15 marks.

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Chapter : 1

Semiconductor diode

Conductor	Semi-conductor	Insulator
		
1. Valence band and conduction band are overlap	1. There is a small energy gap (forbidden gap) bet ⁿ valence band and conduction band.	1. There is large energy gap bet ⁿ valence band and conduction band.
2. It easily conducts electric current.	2. It conducts electric current less than conductor and greater than insulator	2. It doesnot conduct any current.
3. Resistance is very small.	3. Resistance is high.	3. Resistance is very high.
4. Conductor is formed using metallic bonding.	4. Semi-conductor is formed due to covalent bonding.	4. Insulator is formed due to ionic bonding.
5. It has positive temp ^r coefficient.	5. It has negative temp ^r coefficient.	5. It has negative temp ^r coefficient.
6. It has only one valence electron in its outermost orbit	6. It has four valence electrons in its outermost orbit.	6. It has eight valence electrons in its outermost orbit.

eg:- copper,
aluminium etc.

eg:- ~~Si~~ Silicon,
germanium etc.

eg:- paper
wood etc.

* Energy level :- The electrons moving in a particular orbit possess the energy of that orbit. Larger the orbit greater is its energy. In case of single isolated atom, the electrons in any orbit possess definite energy.

* Energy band :- The range of energy possessed by an electron in a solid is known as energy band.

↳ Valence band :- The range of energy possessed by valence electron is known as valence band.
The electrons in the outermost orbit of an atom are known as valence electrons.

↳ Conduction band :- The range of energy possessed by conduction band electrons is known as conduction band.

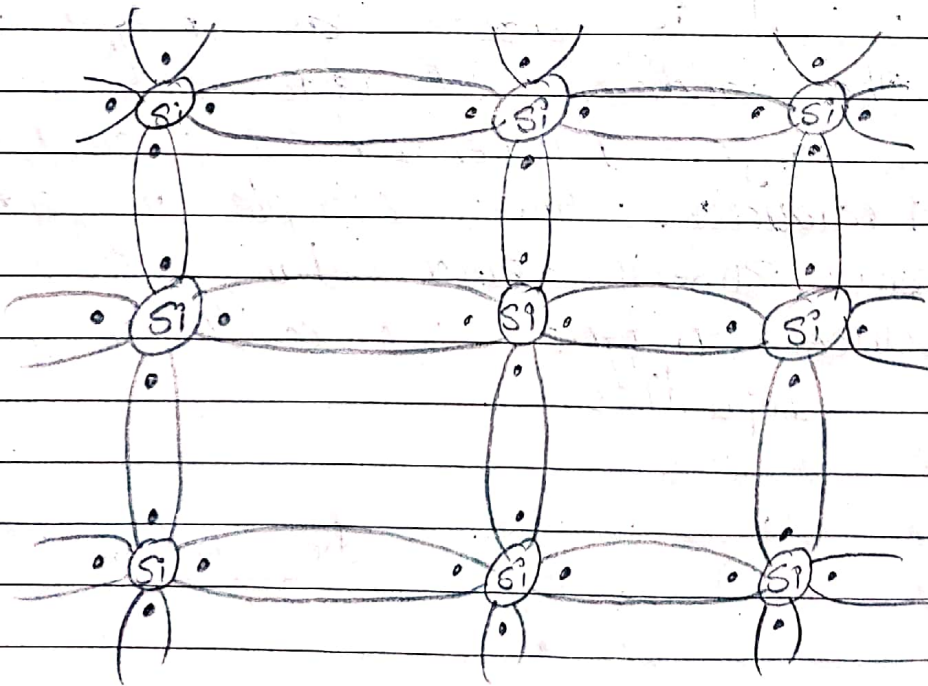
Semi-conductor :-

7 marks Types of semi conductor :-

- 1) Intrinsic semi-conductor
- 2) Extrinsic semi-conductor.

1) Intrinsic Semiconductor:- An extremely pure form semiconductor is known as intrinsic semi-conductor. for egr:- silicon, germanium.

In an intrinsic semiconductor, even at room temp^r, hole-electron pairs are created. When electric field is applied across intrinsic semi-conductor, the current conduction takes place by free electrons and holes. The free electrons are produced due to breaking of some covalent bond by the thermal energy. At the same time holes are created in the covalent bond. Under the influence of electric field, conduction through the semi-conductor is ~~both~~ by both free electrons and holes.



2) Extrinsic Semiconductors :-

When some impurities are added in pure semi-conductor then this forms semi-conductor is known as Extrinsic semi-conductor. The process of adding impurities in semi-conductor is known as doping. Depending upon type of impurities added, extrinsic semi-conductor are classified into :-

- i) n-type semi-conductor
- ii) p-type semi-conductor

i) n-type semi-conductor :- When small amount of pentavalent impurities is added to a pure semi-conductor, it is known as n-type semi-conductor. The addition of pentavalent impurities provides a large number of free electrons in the semi-conductor. The example of pentavalent impurities are arsenic and antimony. Such impurities which produce n-type semi-conductor are known as donor impurities.

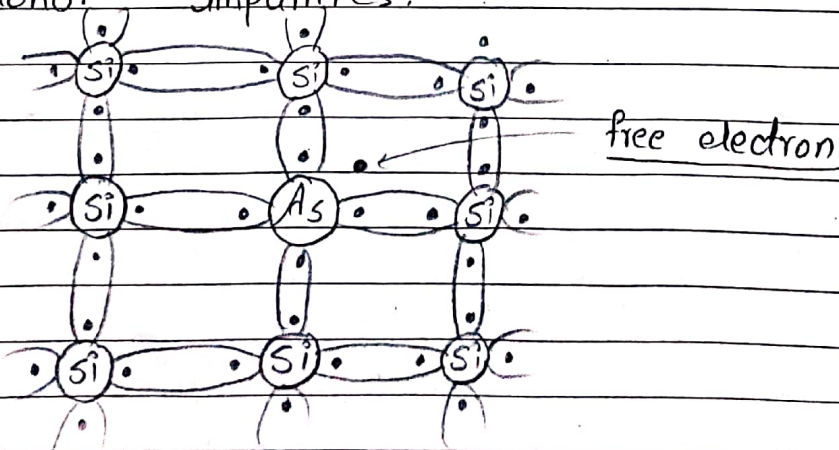


fig:- n-type Semi-conductor

~~n-type~~ ~~semiconductor~~ :- When pentavalent atom (arsenic) is added in pure semiconductor, four valence electrons forms covalent bond with four silicon atoms. The fifth valence electrons of arsenic finds no-place in covalent bond. Therefore the fifth valence electron is free in the silicon crystal. For each arsenic atom added, one free electron will be available in the crystal.

The current conduction in an n-type semiconductor is predominantly by free electrons, i.e. -ve charges.

p-type semiconductor :- When small amount of trivalent impurities is added to the pure semiconductor it is called p-type semiconductor. The addition of trivalent impurities provides the large number of holes in the semi-conductors.

Example of trivalent impurities are gallium and indium.

Such impurities which produce p-type semi-conductors are known as acceptor impurities.

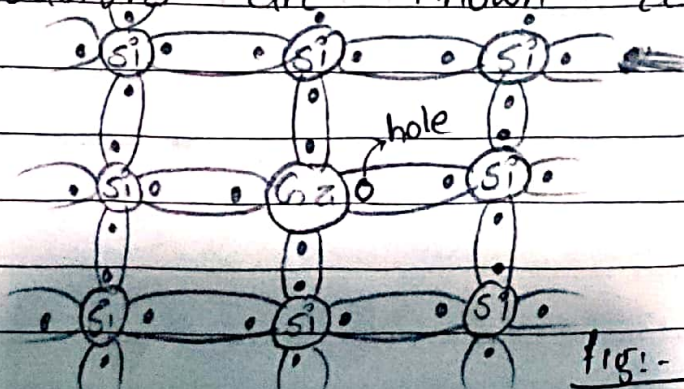


fig:- P-type semiconductor

When trivalent atom of (gallium) is added in pure semi-conductor, three valence electrons forms covalent bond with three silicon atoms. Three valence electrons of gallium can form only three single covalent bond with three silicon atoms. In the fourth covalent bond only silicon atom contributes one valence electron while gallium has no valence electron to contribute. The missing electron is called hole. Therefore, for each gallium atom added, one hole is created.

The current conduction in p-type semi-conductor is pre-dominantly by holes. i.e. +ve charges.

Majority and Minority charge carriers

i) n-types semi-conductor

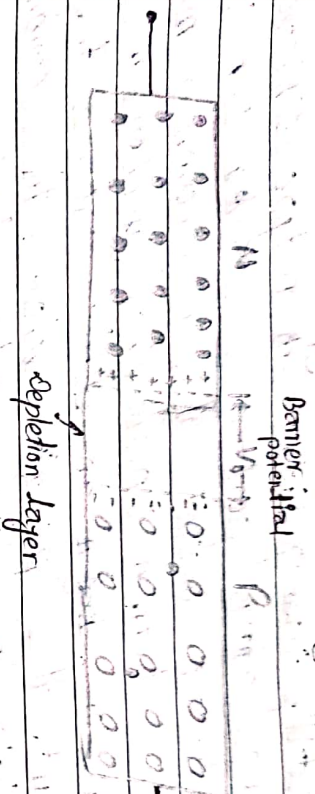
Majority charge carriers :- electrons
Minority charge carriers :- holes

ii) p-types semiconductor

Majority charge carriers :- holes
Minority charge carriers :- electrons

P-n junction :- When p-type of semiconductor is suitably suitably join with n type semiconductor, the contact surface is called P-n junction. At the instant of p-n junction formation, the free electrons near the junction in the n region begin to diffuse across the junction into the p-region where they combine with holes near the junction. Here n-region losses free electron as they diffuse into the junction. This creates the layer of positive charges near the junction. As the electron moves across the junction, the p-region losses holes as the electrons and holes combines, this results a layer of negative charge near the junction in p-region. This two layer of positive and negative charge forms the depletion layer region. The depletion layer is formed very quickly and is very thin compare to n-region and p-region. Once the p-n junction is formed and depletion layer is created, the diffusion of free electron will stop. i.e. the depletion region acts as a barrier to further movement of free electron, across the junction. The positive and negative charge sets up an electric field and there exists a potential difference across the depletion layer which is known as barrier potential. The typical barrier potential for silicon is

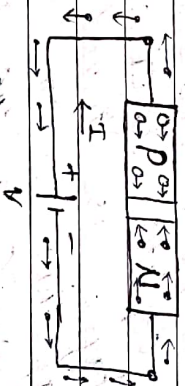
0.7 volt and 0.3 volt for germanium is 0.3 volt.



Biasing of p-n junction :-

→ The process of connecting external dc supply across the p-n junction is known as biasing.

1) Forward Biasing

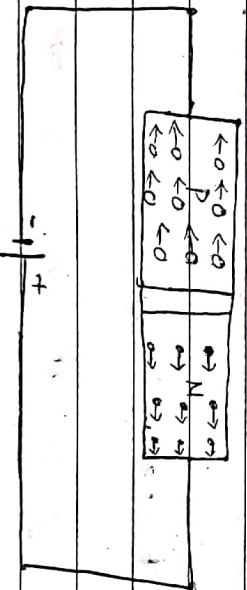


When external dc voltage applied to the junction is in such a direction that it cancels the potential barrier, thus permitting current flow, it is called forward biasing.

When positive terminal of external dc supply is connected to p region and negative terminal of external dc supply is connected to n region then it is called forward biasing.

The applied voltage establish an electric field which acts against the field due to barrier potential... therefore the resultant field is weakened and the barrier height is reduced at the junction. As the barrier voltage is very small, a small forward voltage is sufficient to completely eliminate the barrier. Once the potential barrier is eliminated by the forward voltage, junction resistance becomes almost zero and a low resistive path is established for the entire circuit. Therefore current flows in the circuit which is known as forward current.

2) Reverse Biasing



When positive terminal of external dc supply is connected to n-region and negative terminal is connected to p-region, external dc supply is connected to p-region in such a way that potential barrier is increased then it is known as reverse biasing. The applied reverse voltage establish an electric field which acts in the same direction as the field due to potential barrier. Therefore resultant field at the junction is strengthened and

the barrier height is increased. the increased potential barrier prevents the flow of charge carrier across the junction. thus a heighted resistance path is establish and no current flows in the circuit.

~~P-n junction diode~~
→ ~~then the~~

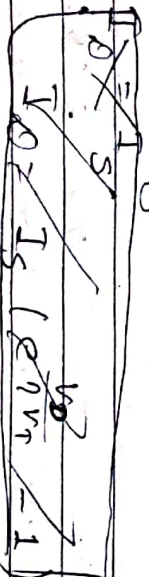
P-n junction diode
→ It is a device formed by suitably joining P-type and n-type semi-conductor. It is unidirectional device and has two terminals named as anode and cathode.

anode  cathode

fig: symbol of P-n junction diode

Anode is a terminal from P-region.
cathode is a terminal from n-region.

Current flowing through diode is given by



$$I_0 = I_s \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

where,

I_s = reverse saturation current

V_D = Voltage across diode

η = intrinsic ratio

≈ 1 for Germanium

≈ 2 for silicon

V_T = Thermal voltage

$$= \frac{kT}{q}$$

k = Boltzmann's const. $= 1.38 \times 10^{-23} \text{ J K}^{-1}$

T = Temperature

q = Charge of electron $= 1.6 \times 10^{-19} \text{ C}$

V-I characteristics of p-n junction diode:-

→ This characteristics of p-n junction diode gives the relation betⁿ voltage across the diode and current through the diode. The characteristics can be studied under zero external voltage forward bias and reverse bias.

i) Zero external voltage

ii) Forward bias

iii) Reverse bias

ii) Zero- External diode

→ When external voltage is zero, voltage across the diode and current through diode is zero which is indicated by point 'O' in the figure.

iii) Forward bias

→ With forward bias to p-n junction diode i.e. anode connected to positive terminal and cathode connected to negative terminal, the potential barrier is reduced. At some forward voltage potential barrier is eliminated and current starts flowing in the circuit. Initially the current increases very slowly and the curve is non-linear which is indicated by region OA in the figure. However, once the external voltage exceeds the potential barrier, the current rises very sharply with increasing external voltage. Now, the curve is almost linear which is indicated by region AB.

iv) Reverse bias

With the reverse bias to p-n junction diode i.e. cathode connected to positive terminal and anode connected to negative terminal, an external dc supply, the potential barrier is increased. Therefore junction resistance becomes very high and practically no current flows through the circuit. However, a very small current flows in the circuit due to minority carriers with reverse

bias as shown in figure. This current is known as reverse saturation current. If reverse voltage is increased continuously the kinetic energy of electron may become high enough to knock out from the semi-conductor atom. At this stage breakdown of junction occurs and reverse current will rise suddenly. This may destroy the junction permanently.

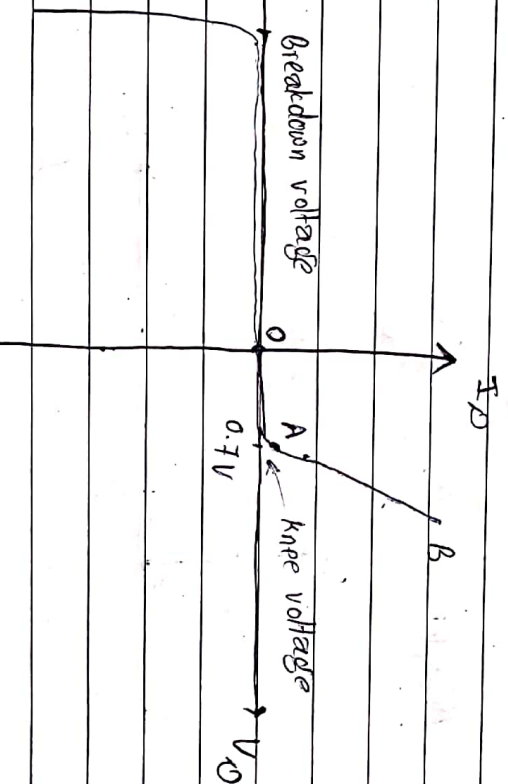


fig: V-I characteristics of p-n junction diode

* Breakdown voltage :- It is minimum reverse voltage at which p-n junction breaks down with sudden rise in reverse current.

* Knee voltage :- It is the forward voltage at which the current through the junction starts to increase rapidly.

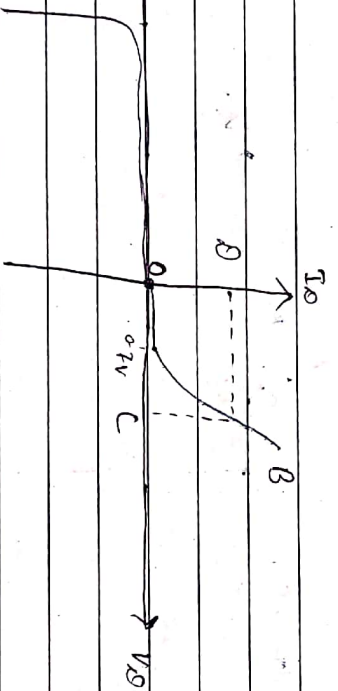
* Resistance of p-n junction diode :- Imp

i) AC resistance (static resistance) :- ocured

It is opposition ^{or offered} offered by diode to the direct current. It is measured by the ratio of the dc voltage across the diode to the ~~the~~ resultant dc current through it.

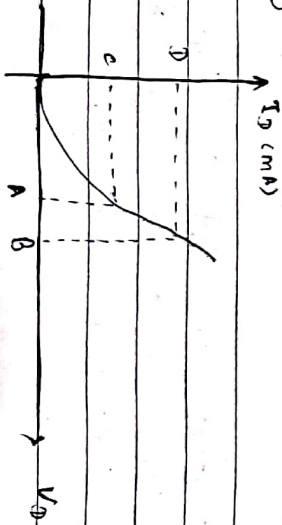
It is given by

$$\text{Resistance } R_f = \frac{OC}{OD}$$



ii) AC resistance (dynamic resistance) :-

It is the opposition offered by diode to the changing forward current. It is measured by the ratio of change in voltage across diode to the resulting change in current.



~~Imp~~ ac resistance (r_f) = $\frac{V_{AC}}{I_{CO}} = \frac{\Delta V_D}{\Delta I_D} = \frac{dV_D}{dI_D}$

We have,

$$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right)$$

diff. w.r.t V_D ,
 $\frac{dI_D}{dV_D} = \frac{d}{dV_D} \left[I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right) \right]$

$$\frac{dI_D}{dV_D} = I_S \frac{d}{dV_D} e^{\frac{V_D}{nV_T}}$$

$$\frac{dI_D}{dV_D} = I_S * \frac{1}{nV_T} e^{\frac{V_D}{nV_T}}$$

$$\frac{1}{r_f} = \frac{1}{nV_T} I_S e^{\frac{V_D}{nV_T}}$$

Since $I_{DQ} \approx I_S e^{\frac{V_D}{nV_T}}$

$$\therefore \frac{1}{r_f} = \frac{I_{DQ}}{nV_T}$$

$$\therefore r_f = \frac{nV_T}{I_{DQ}} \quad \text{--- (*)}$$

Effect of Temperature on V-I characteristics

The reverse saturation current I_s approximately doubles for every 10°C rise in temperature for silicon. If I_{s1} be the reverse saturation current at temperature T_1 , then at any temp (T) , it is given by

$$I_s = I_{s1} \times 2^{\left(\frac{T-T_1}{10}\right)}$$

The ~~V-I~~ V-I characteristics curve for silicon diode at various temp^r as shown in figure. The forward diode current (I_F) increases with increase in temp^r. In reverse biasing more covalent bonds break down at higher temp^r. As a result more electrons holes pair are generated thus leakage saturation current also increases. The junction breakdown occurs at lower reverse voltage at ~~higher~~ higher temp^r.

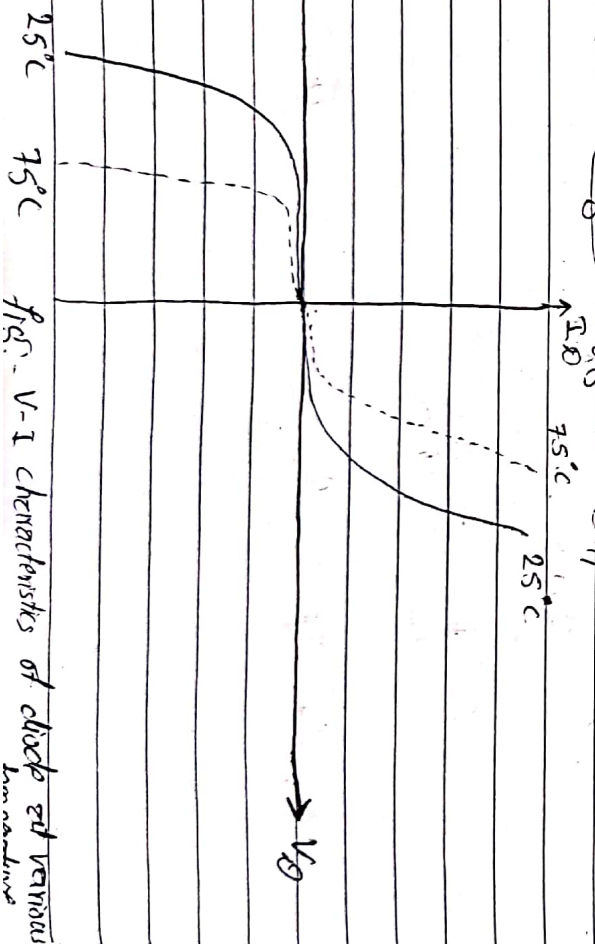


Fig. V-I characteristics of diode at various temperatures

Value of $\eta \Rightarrow$ $G \rightarrow 1$ Germanium $\rightarrow 1$
 $S \rightarrow 2$ Silicon $\rightarrow 2$

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Q Current flowing through a silicon diode at room temp is 10 μ Amp. When large reverse voltage is applied calculate the current through diode when 0.5 V forward voltage is applied.

Solution

Given

Reverse saturation current (I_s) = 10 μ A

Forward voltage (V_o) = 0.5 V

diode current (I_o) = ?

We have,

$$I_o = I_s \left(e^{\frac{V_o}{nV_T}} - 1 \right)$$

$$V_T = \frac{kT}{q}$$

Room temperature = 25 $^{\circ}$ C

At room temp,

$$V_T = \frac{kT}{q} \approx 26 \text{ mV}$$

When temp is room temp
[const. value] Put $V_T = 26 \text{ mV}$

~~$$I_o = I_s \left(e^{\frac{V_o}{nV_T}} - 1 \right)$$~~

$$\therefore I_o = 10 \times 10^{-6} \left(e^{\frac{0.5}{2 \times 26 \times 10^{-3}}} - 1 \right)$$

$$= 0.1499 \text{ A}$$

$$= 149.92 \text{ mA}$$

$$k = \text{boltzmann const} = 1.38 \times 10^{-23}$$

$$q = 1.6 \times 10^{-19}$$

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Q Find the leakage saturation current at 100°C for a silicon diode whose leakage saturation current at 27°C is $10 \mu\text{A}$.
solution

Reverse saturation current at 27°C (I_{S1}) = $10 \mu\text{A}$

Reverse saturation current at 100°C (I_{S2}) = ?

$$I_{S2} = I_{S1} \times 2^{\frac{T_2 - T_1}{T_0}} \rightarrow \left(\frac{100 - 27}{10} \right)$$

$$= 1575.86 \times 10^{-3} \text{ A}$$

$$= 1.57 \text{ mA}$$

Q A silicon diode has a reverse sat. current of $10 \mu\text{A}$ at 20°C . Find its current when it is forward biased by 0.55 V . find the current in the same diode when temp rises to 100°C .

Here

Reverse saturation current at 20°C = $10 \mu\text{A}$

(silicon diode) = ?

forward voltage (V_0) = 0.55

Current (I_0) = ?

at 20°C
Diode current (I_0) = $I_S \left(e^{\frac{V_0}{nV_T}} - 1 \right)$

$$V_T = \frac{kT}{q} = \frac{1.38 \times 10^{-23} \times (273 + 20)}{1.6 \times 10^{-19}}$$

$$= 0.0257$$

Now,

$$I_D = 10 \times 10^{-6} \left(e^{\frac{0.55}{2 \times 0.025}} - 1 \right)$$

$$= 0.598 \text{ Amp} \#$$

Again, at 100°C

$$\text{Value of } I_S = I_{S1} \times 2^{\left(\frac{T - T_1}{10} \right)} \left(\frac{100 - 20}{10} \right)$$

$$= 10 \times 10^{-6} \times 2$$

$$= 0.00256$$

$$= 2.56 \times 10^{-3} \text{ Amp}$$

$$V_T = \frac{kT}{q} = \frac{1.38 \times 10^{-23} \times (100 + 273)}{1.6 \times 10^{-19}}$$

$$= 0.0321 \text{ V}$$

$$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right)$$

$$= 2.56 \times 10^{-3} \left(e^{\frac{0.55}{2 \times 0.0321}} - 1 \right)$$

$$= 13.45 \text{ amp}$$

Q

A silicon diode carries a current of 10 mA when forward bias of 0.2 V is applied at room temperature

a) Calculate reverse saturation current

b) Calculate diode voltage needed for diode current 1 mA and 100 mA.

c) Calculate reverse saturation current at 20°C above

room temperature and also find diode current at 0.2V forward bias.

Here

$$I_D = 10 \text{ mA} = 10 \times 10^{-3} \text{ A}$$

$$V_D = 0.2 \text{ V}$$

$$I_S = ?$$

(a)

$$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right)$$

At room temp,

$$V_T = 26 \text{ mV}$$

$$I_S = \frac{I_D}{\left(e^{\frac{V_D}{nV_T}} - 1 \right)}$$

$$I_S = \frac{10 \times 10^{-3}}{\left(e^{\frac{0.2}{1 \times 26 \times 10^{-3}}} - 1 \right)}$$

$$= 0.000218$$

$$= 2.18 \times 10^{-4} \text{ amp}$$

(b)

For $I_D = 1 \text{ mA}$

$$V_D = ?$$

$$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right) \quad [\text{neglect } 1]$$

$$1000 I_S = e^{\frac{V_D}{nV_T}}$$

$$\frac{I_D}{I_S} = e^{\frac{V_D}{nV_T}}$$

Taking log on either side,

$$\ln \left(\frac{I_D}{I_S} \right) = \ln e^{\frac{V_D}{nV_T}}$$

$$\ln \left(\frac{I_D}{I_S} \right) = \frac{V_D}{nV_T}$$

$$V_D = nV_T \ln \left(\frac{I_D}{I_S} \right)$$

$$V_D = 2 \times 26 \text{ mV} \ln \left(\frac{1}{2.18 \times 10^{-4}} \right)$$

$$= 0.438$$

For $I_D = 100 \text{ mA}$

c)

$$T - T_1 = 20^\circ\text{C}$$

$$I_S = I_{S_1} \times 2^{\frac{20}{10}}$$

$$= 2.18 \times 10^{-4} \times 2^{\frac{20}{10}} = 4$$

$$V_T = \frac{kT}{q} \Rightarrow V_T = 26 \text{ mV}$$

let room temp $\approx 27^\circ\text{C}$

$$V_T = \frac{1.38 \times 10^{-23} \times (27 + 273)}{1.6 \times 10^{-19}}$$

$V_T = 26 \text{ mV}$
 room temp $= 27^\circ\text{C}$
 us 52 mV

$$= 0.025 \text{ mV}$$

now,

$$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right)$$

$$= 4 \left(e^{\frac{0.2}{2 \times 0.025}} - 1 \right)$$

$$= 214.39$$

Junction capacitance

→ The depletion region across the junction behaves as an insulating material or dielectric constant of capacitor and p and n sides acts as parallel plates. Thus junction exhibits the property of capacitance which is called junction capacitance. There are two types of junction capacitance. They are

- i) Diffusion capacitance
- ii) Transition capacitance.

i → Diffusion capacitance :- It occurs when diode is forward biased. With forward biasing the depletion width reduces, so diffusion capacitance increases. Diffusion capacitance is directly proportional to the forward diode current I_0 .

$$C_D = \frac{dQ}{dV}$$

$$C_D = \frac{I_0 \tau}{V_0} \left[\frac{dI}{dV} \right]$$

$$= C \cdot d \left(I_0 e^{\frac{V_0}{nV_T}} - 1 \right)$$

$$= C \cdot I_0 \cdot \frac{1}{nV_T} e^{\frac{V_0}{nV_T}}$$

$$= \frac{C}{nV_T} I_0 e^{\frac{V_0}{nV_T}}$$

Since, $I_D \approx I_S e^{\frac{V_D}{nV_T}}$

$$\therefore C_D = \frac{C}{nV_T} \times I_D \quad \text{--- (*)}$$

So, $C_D \propto I_D$

2) Transition Capacitance (C_T)

→ It occurs when the diode is reversed biased.
With reverse biasing, the depletion width increases so, transition capacitance will reduce. The depletion width can be written as

$$W^2 = \frac{2\varepsilon V_D}{e} \left[\frac{1}{N_A} + \frac{1}{N_D} \right]$$

where,

N_A = Number of acceptor atoms

N_D = Number of donor atoms

V_D = Applied voltage across junction.

e = charge of an electron.

As,

$$C_T = \frac{\varepsilon A}{W}$$

$$= \frac{\varepsilon A}{\sqrt{\frac{2\varepsilon V_D}{e} \left[\frac{1}{N_A} + \frac{1}{N_D} \right]}}$$

$$= \frac{\epsilon A}{\sqrt{\frac{2\epsilon V_A}{e} \left[\frac{N_A + N_D}{N_A N_D} \right]}}$$

$$\therefore C_T \propto \frac{1}{\sqrt{V_A}}$$

Thus, transition capacitance is inversely proportional to square root of applied reverse voltage.

II Junction Breakdown :

In reverse biasing, a very small current flows due to minority charge carriers. This current is known as leakage saturation current.

If reverse voltage is further increased the large current flows suddenly at a certain reverse bias voltage. This mechanism of producing a large reverse current is called breakdown.

There are two breakdown mechanisms. They are:

- i) Zener breakdown
- ii) ~~Avalanche~~ Avalanche breakdown

i) Zener breakdown

→ It occurs due to high electric field intensity across the junction. It occurs normally at low reverse voltage. Zener breakdown occurs in a

heavily doped diode. Zener breakdown voltage decreases with rise of temperature.

ii) Avalanche breakdown

→ It occurs due to a collision of charge carriers with other atoms. It occurs at a high reverse voltage. Avalanche breakdown occurs in lightly doped diode. Avalanche breakdown voltage increases with increase of temperature.

Equivalent Circuit of p-n junction diode :



Fig: ideal diode



ideal diode

Fig: Approximate model of p-n junction diode.

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Zener diode

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→ A properly doped crystal diode which has a sharp breakdown voltage is known as zener diode. A zener diode is like an ordinary diode except that it is properly doped so that as to have a sharp breakdown voltage. A zener diode is always reverse connected i.e. always reversed biased. Zener diode has a sharp breakdown voltage which is known as zener voltage. With forward biasing, its characteristics is just same as ordinary diode.

Symbol of zener diode :-

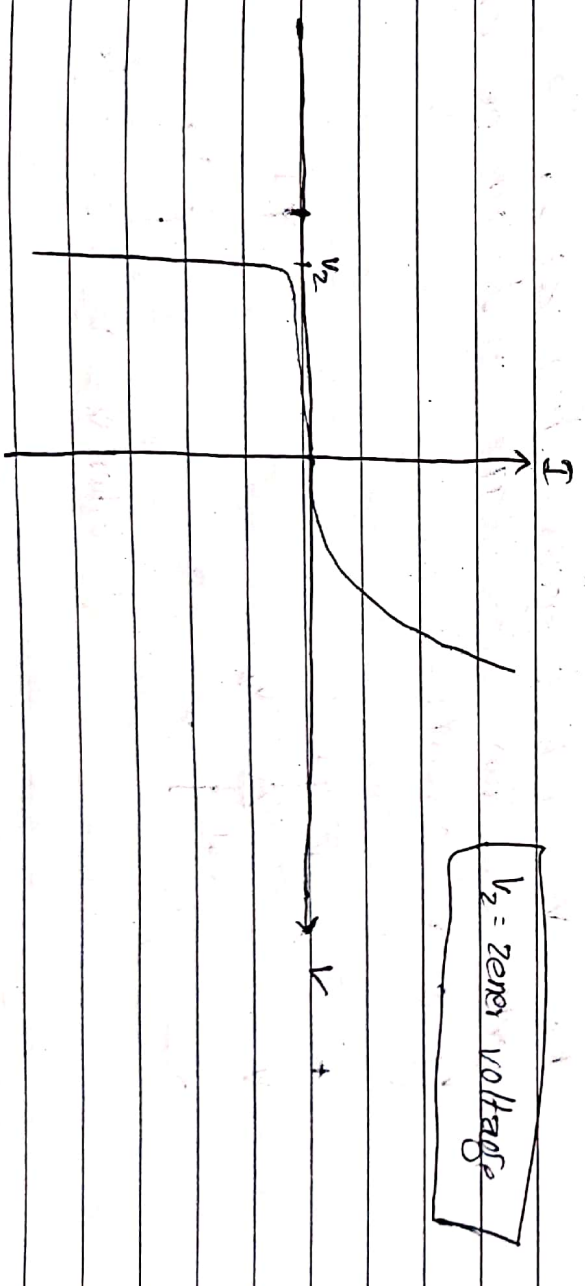
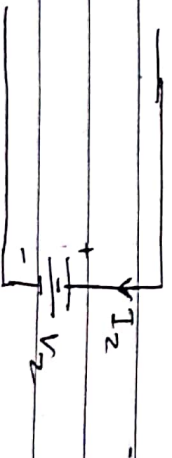
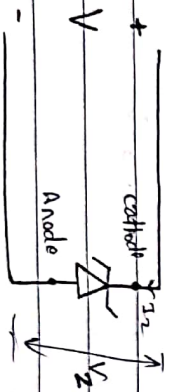


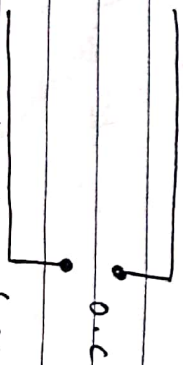
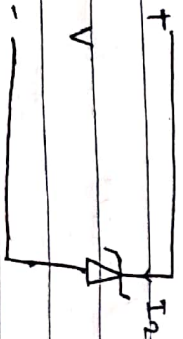
Fig.:- V-I characteristics of zener diode

When reverse voltage across a zener diode is equals to or more than breakdown voltage (V_z), the current increases very sharply. In this region the curve is almost vertical. It means that the voltage across zener diode is constant at V_z . When the reverse voltage across the zener diode is less than V_z but greater than zero volt, the zener diode is in the off state.

When reverse voltage across zener diode is equals to or greater than V_z , voltage across zener diode is constant and can be represented by a battery of voltage V_z . When reverse voltage across zener diode is less than V_z but greater than zero, the zener diode is in off state and can be represented by open circuit.

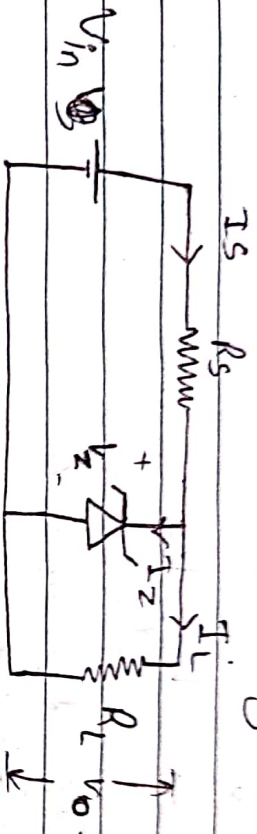


When $V \geq V_z$, ON state



When $0 \leq V < V_z$ 'off state'

Imp Zener diode as voltage regulator :-



Let us consider a circuit consisting of a zener diode with zener voltage V_Z connected across load R_L ~~series~~ as shown in figure. The series resistance R_S absorbs the output that fluctuation so as to maintain constant voltage across the load. The zener will maintain a constant voltage V_Z across the load so long as the input voltage does not fall below V_Z .

Suppose the input voltage increases. Since the zener is in breakdown region, ~~the~~ the output voltage remains constant. ~~The~~ The excess voltage is dropped across the series resistance R_S . This will cause an increase in current I_S . The zener will conduct increase of current in I_S while the load current remains constant. Hence output voltage V_o remains constant irrespective of change in input voltage.

$$V_0 \approx I_L R_L$$

$$I_S = \frac{V_{in} - V_Z}{R_S}$$

$$-I_S + I_Z + I_L \stackrel{!}{=} 0$$

$$I_S = I_Z + I_L$$

Suppose the input voltage is constant but the load resistance (R_L) decreases. This will cause an increase in load current (I_L). The extra current cannot come from the source. The additional current will come from decrease in zener current (I_Z). Consequently, the output voltage ~~will~~ stay at constant voltage.

$$V_0 \approx I_L R_L$$

$$I_S = \frac{V_{in} - V_Z}{R_S} \quad \begin{matrix} \rightarrow \text{const} \\ \rightarrow \text{const} \end{matrix}$$

$$I_S \approx I_Z + I_L \quad \uparrow$$

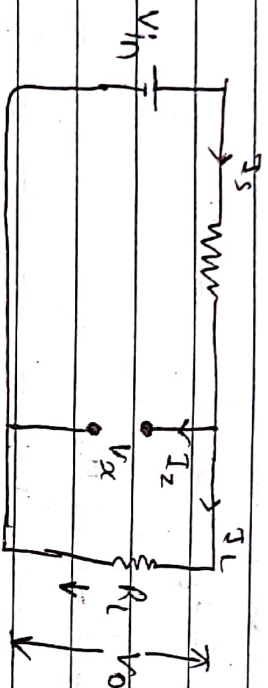
Solving voltage regulator circuits:

Check either the zener diode is in breakdown region or not.

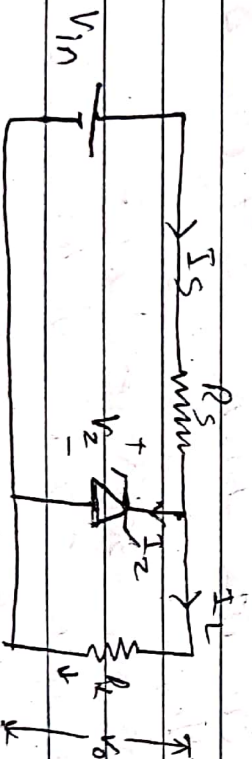
$$V_Z = \frac{V_{in} * R_L}{R_S + R_L}$$

Q8 If $V_x \geq V_z$ then Zener diode is in breakdown region or in ON state and voltage across Zener diode will remain constant (i.e. V_z)

If $V_x < V_z$, then Zener diode will be in off state and acts as open circuit.



Q9 Use following relations:



When Zener is in ON state

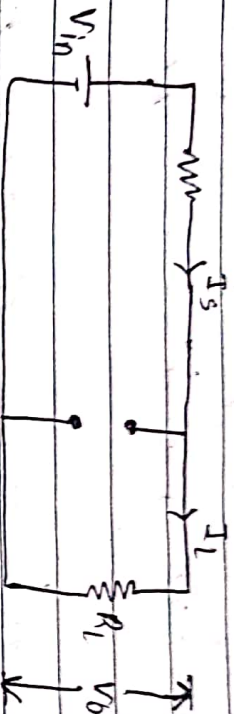
$$V_o = V_z$$

$$I_L = \frac{V_o}{R_L} = \frac{V_z}{R_L}$$

$$I_S = \frac{V_{in} - V_z}{R_S}$$

$$I_S = I_L + I_Z$$

When zener is in off state,



$$I_Z = 0$$

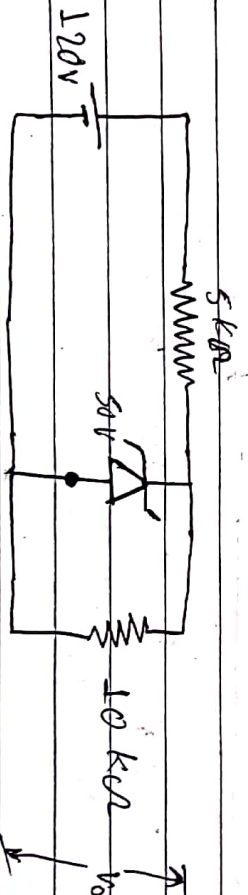
$$I_S = I_L$$

$$V_o = I_L \times R_L$$

$$I_L = \frac{V_{in}}{R_s + R_L}$$

Q for the circuit shown in figure below find
i) output voltage

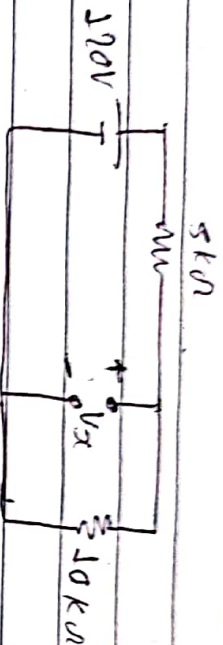
- ii) voltage across series resistance
- iii) current through zener diode.



Solution

Step 1:- find voltage across zener diode to check either the zener diode to check either the

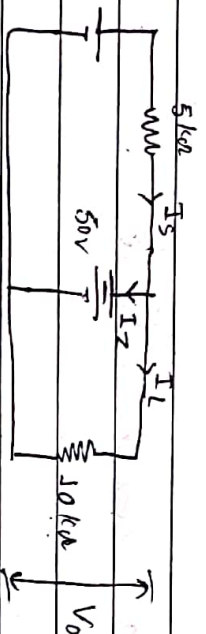
Zener diode I_s in on state & or off state.



$$V_x = \frac{V_{in} \times R_L}{R_L + R_S} = \frac{120 \times 10}{5 + 10} = 80 \text{ V}$$

Since, $V_x > V_z$, so Zener is in on state.

(i) $\therefore$ output voltage (V_o) = $V_z = 50 \text{ V}$



$$\begin{aligned} \text{(ii)} \quad \text{Voltage across series resistance} &= V_{in} - V_z \\ &= 120 - 50 \\ &= 70 \text{ V} \end{aligned}$$

$$\text{(iii)} \quad I_s = I_L + I_z$$

$$I_z = I_s - I_L$$

$$I_L = \frac{V_o}{R_L} = \frac{V_z}{R_L} = \frac{50}{10000} = 5 \text{ mA}$$

$$I_S = \frac{V_{in} - V_Z}{R_S}$$

$$= \frac{120 - 50}{5000}$$

$$= 14 \text{ mA}$$

$$\therefore I_Z = I_S - I_L$$

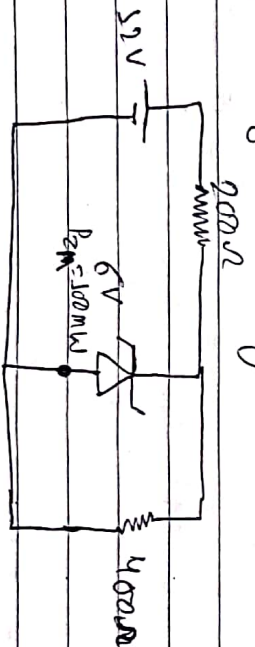
$$= 14 - 5$$

$$= 9 \text{ mA}$$

Q for the circuit shown in figure below find

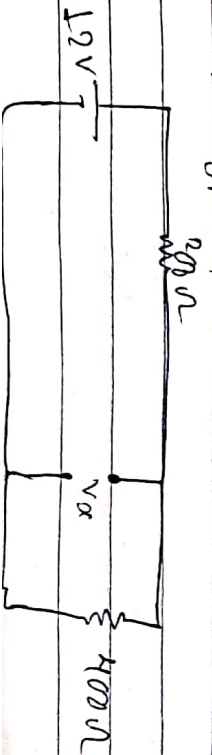
- i) Output voltage
- ii) Current through zener diode
- iii) Zener power

Determine whether the zener diode is operating safely or not.



Solⁿ ~~1~~ checking whether the zener diode

is on or off.



$$V_x = \frac{V_{in} \times R_L}{R_L + R_S}$$

$$= \frac{12 \times 400}{200 + 400}$$

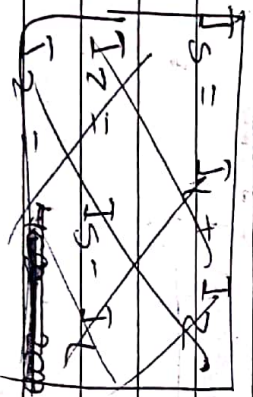
$$= \cancel{0.25} \text{ } 8 \text{ V}$$

Since, $V_a > V_z$, ^{S_0} since Zener is on state.

is output voltage (V_o) = $V_z = 6$

ii) Current through Zener diode,

$$I_Z = \frac{V_o - V_Z}{R_L} = \frac{6 - 400}{400} = 0.015$$



$$I_S = \frac{V_{in} - V_Z}{R_S}$$

$$= \frac{12 - 6}{200}$$

$$= 0.03 \text{ amp}$$

Now,

$$I_S = I_L + I_Z$$

$$I_Z = I_S - I_L$$

$$= 0.03 - 0.015$$

$$= 0.015 \text{ amp.}$$

Now,

$$P_Z = V_Z \times I_Z$$

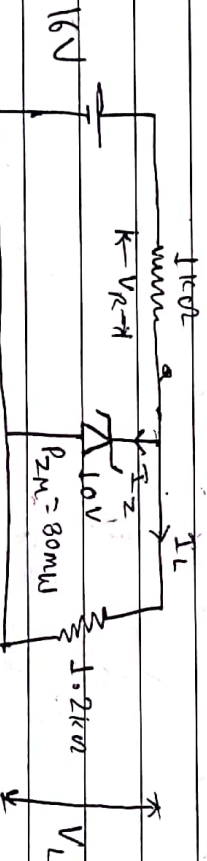
$$= 6 \times 0.015$$

$$= 0.09 \text{ watt}$$

$$= 90 \text{ mW}$$

since, $P_Z < P_{Z_{max}}$, so Zener is operating safely

Q For the zener diode network shown in figure determine V_L , V_R , I_Z and P_Z .



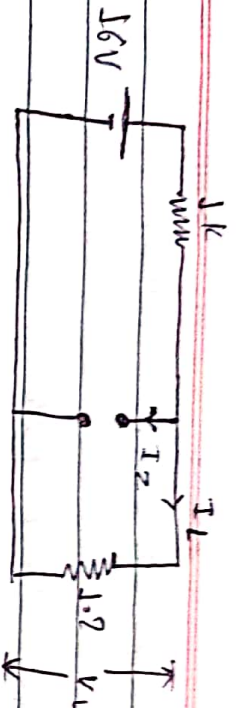
How

$$V_R = \frac{V_{in} \times R_L}{R_L + R_s}$$

$$= \frac{16 \times 1.2k}{1.2k + 1k}$$

$$= 8.72$$

Since, $V_R < V_Z$ so the zener diode is OFF state.



$$V_L = I_L \cdot R_L$$

$$= \frac{V_{in}}{R_S + R_L} \times R_L$$

$$= \frac{16}{1 + 1.2} \times 1.2$$

$$= 8.72$$

$$I_L = \frac{V_{in}}{R_S + R_L}$$

$$= \frac{16}{1 + 1.2}$$

$$= 7.27 \text{ mA}$$

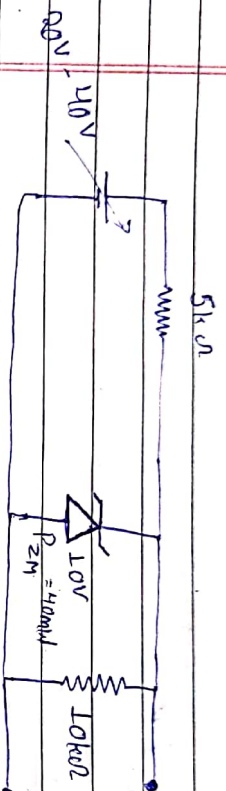
$I_2 = 0 \text{ amp}$ [∵ open circuit]

$$P_2 = V_2 \times I_2$$

$$= 10 \times 0$$

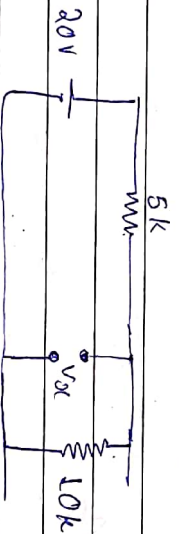
$$= 0 \text{ watt}$$

- Q for the circuit given below, find,
- Output voltage
 - maximum current to zener diode I_{zm}
 - minimum current to zener diode I_{zm}



Solution

i) Check either Zener diode is on or not,

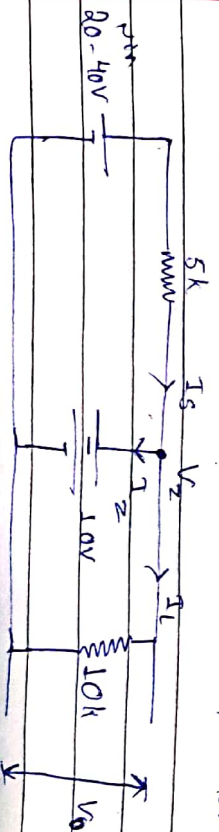


$$V_x = \frac{V_{in} \times R_L}{R_S + R_L}$$

$$= \frac{20 \times 10}{5 + 10}$$

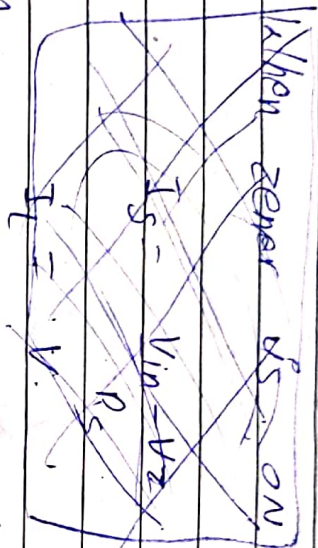
$$= 13.33 \text{ V}$$

Since, $V_x < V_z$, $V_x > V_z$, Zener is in on state



$$\therefore V_0 - V_Z = 10V$$

$$I_L = \frac{V_0}{R_L} = \frac{10}{10} = 1 \text{ mA}$$



As

$$I_Z = I_S - I_L \quad \text{constant}$$



$$I_{Z \min} = I_{S \min} - I_L$$

$$I_{Z \max} = I_{S \max} - I_L$$

Since

$$I_S = \frac{V_{in} - V_Z}{R_S} \quad \text{const.}$$

$R_S \rightarrow \text{const}$

$$\therefore I_{S \max} = \frac{V_{in \max} - V_Z}{R_S}$$

$$\Rightarrow 0.002 \text{ amp}$$

$$= \frac{40 - 10}{5000}$$

$$\therefore I_{Z \max} = I_{S \max} - I_L$$

$$= 0.006 \text{ amp}$$

$$= 6 \text{ mA}$$

$$= 6 \text{ mA}$$

$$= 5 \text{ mA}$$

$$\therefore I_{S \min} = \frac{V_{in \min} - V_Z}{R_S}$$

$$\therefore I_{Z \min} = I_{S \min} - I_L$$

$$= \frac{20 - 10}{5000}$$

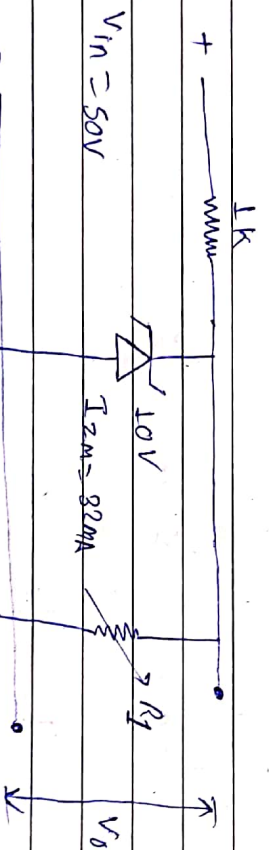
$$= 2 \text{ mA}$$

$$P_{Z_{max}} = 5 \times 10$$

$$= 50 \text{ mW}$$

Since $P_{Z_{max}} > P_{ZM}$ so Zener diode is not operating safely

Q For the circuit shown in fig below determine the range of R_L and I_L that results the Zener diode in conducting state.



Zener is in ON state (given in question)

$$\therefore V_o = V_Z = 10V$$

$$I_L = \frac{V_o}{R_L} \rightarrow \text{const}$$

$$I_{L_{min}} = \frac{V_o}{R_{L_{max}}}$$

$$I_{L \max} = \frac{V_0}{R_{L \min}} \quad \left[\because I_S = \frac{V_{in} - V_Z}{R_S} \right] \quad \text{const. because circuit is ON.}$$

$$I_L = I_S - I_Z \quad \text{const.}$$

$$I_S = \frac{V_{in} - V_Z}{R_S}$$

$$= \frac{50 - 10}{1000}$$

$$= 40 \text{ mA}$$

$$I_{L \max} = I_S - I_{Z \min}$$

$$= 40 - 0$$

$$= 40 \text{ mA}$$

$$I_{L \min} = I_S - I_{Z \max}$$

$$= 40 - 32$$

$$= 8 \text{ mA}$$

$$\therefore R_{L \max} = \frac{V_0}{I_{L \min}}$$

$$= \frac{10}{8 \times 10^{-3}}$$

$$= 1250 \Omega$$

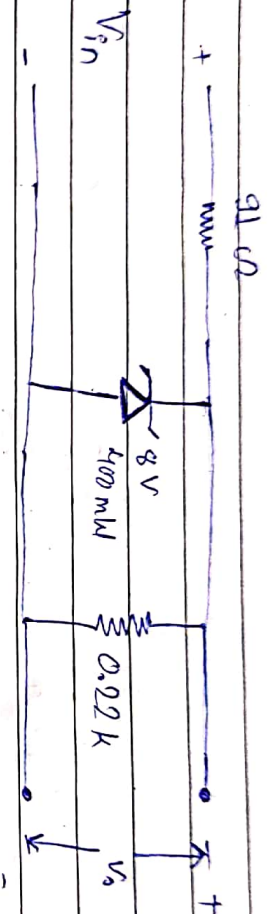
$$= 1.25 \text{ k}\Omega$$

$$\therefore R_{L \min} = \frac{V_0}{I_{L \max}}$$

$$= \frac{10}{40 \times 10^{-3}}$$

$$= 250 \Omega$$

Q Determine, the range of input voltage that will maintain load voltage at 8 volt without exceeding the maximum power rating of zener diode.



~~Zener is not in ON state~~
Zener is in ON state,

$$V_0 = V_Z = 8 \text{ V}$$

$$I_L = \frac{V_0}{R_L} = \frac{8}{0.22} = 36.36 \text{ mA} \quad \text{const.}$$

$$I_S = \frac{V_{in} - V_Z}{R_S} \quad \text{const.}$$

$$I_S = I_L + I_Z$$

$$V_{oc} = \frac{V_{in} \times R_L}{R_L + R_S}$$

When $V_{oc} = V_Z$ then, $V_{in} = V_{in \text{ min}}$

$$V_{in \text{ min}} = \frac{I_S \times R_S + V_Z}{I_S} = \frac{I_S \times R_S + V_Z}{I_S} \quad \text{const.}$$

$$8 = \frac{V_{in} \times 0.22}{0.22 + 91}$$

$$8 = \frac{V_{in} \times 220}{220 + 91}$$

$$V_{in \min} = 11.81 \text{ V}$$

Again,

$$I_{s \max} = I_L + I_{2 \max}$$

Since,

$$P_{2 \max} = V_2 \times I_{2 \max}$$

$$I_{2 \max} = \frac{P_{2 \max}}{V_2} = \frac{400 \times 10^{-3}}{8} = 50 \text{ mA}$$

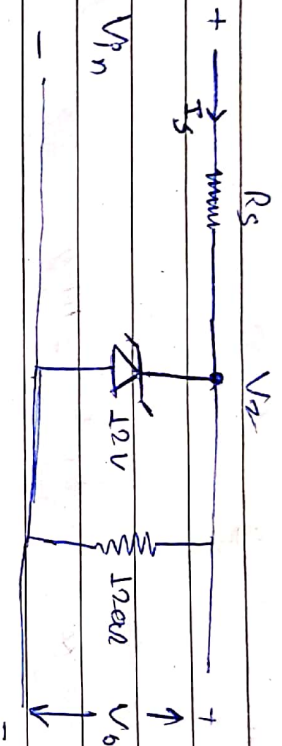
$$I_{s \max} = 86.36 + 50 = 136.36 \text{ mA}$$

$$\therefore V_{in \max} = I_{s \max} \times R_S + V_Z$$

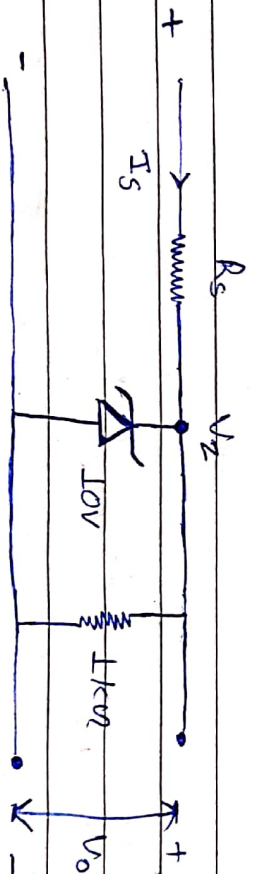
$$= 86.36 \times 10^{-3} \times 91 + 8$$

$$= 15.85 \text{ V}$$

Q The zener diode neglect regulator shown in figure has a fix voltage of 12 V across load as long as I_Z is maintained betⁿ 20 mA and 200 mA. Find R_S so that V_L remains 12 V while V_{in} varies from 15 V to 19.5 V,



Q For the given zener diode regulator circuit, input voltage varies from 18 volt to 22 volt. What should be the value of resistance R_S to operate the zener diode as a regulator circuit. The minimum zener current must be 5 mA to insure that the zener diode works as a regulator.



Diode Switching time

The electrical can be made on and off by forward biasing or reverse biasing the diode. In both case the diode response as transient and an interval of time elapses before the diode reaches its steady state. The time interval for the recovery of steady state of the diode is called the recovery time.

not in
mult

Forward recovery time :- On switching over a diode from reverse biased condition to forward biased condition, the diode takes certain time interval called forward recovery time. It is defined as the time interval betⁿ the instant of 10% of diode voltage to the 90% of its final value.

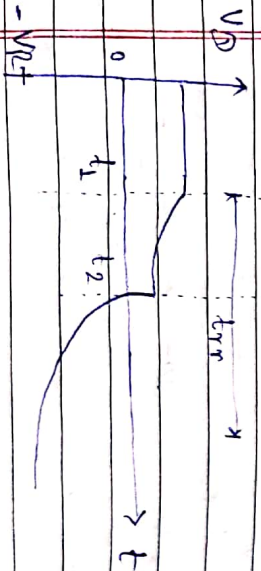
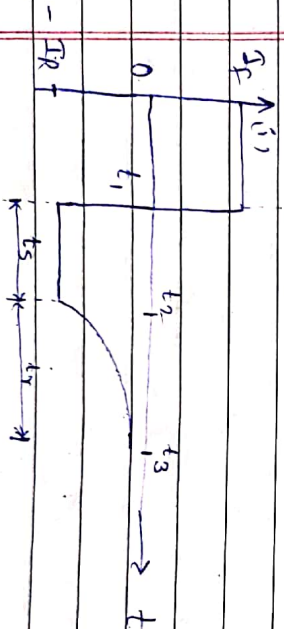
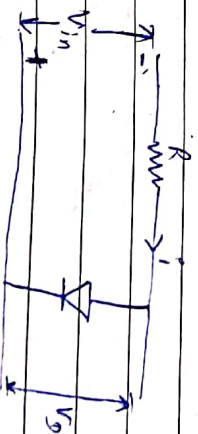
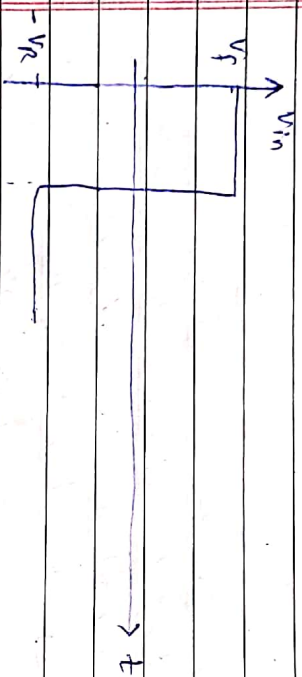
* Reverse recovery time :- When p-n junction is forward biased, a large number of electrons cross over from n region to p region and large number of holes cross over from the p region to n region. The electrons in p region and holes in n region establish a large number of minority carriers in each region. The number of minority carrier is very large near to the junction and decreases as the distance from the junction increases. ~~For~~

If the external voltage is suddenly

t_{rr} = reverse recovery time
 t_s = storage time
 t_r = transition time

Date _____
 Page _____

reversed, the diode ^{current} will not immediately falls to its steady state reverse voltage value. However, due to large number of minority carriers in both region, the diode ~~recovered~~ will simply reverse and stay at measurable level. The diode will continue to conduct for a certain time which is known as reverse recovery time.



At $t = t_1$, input voltage suddenly reverses to $-V_r$. The current does not drop to zero but maintained at certain value for the $-T_r$. For the time duration t_{sr} (storage time) required for the minority carriers to return their majority carrier state in the opposite region.

At $t = t_2$, the process of sweeping of the excess minority carriers across the junction has completed. So, the diode voltage begins to reverse and the magnitude of diode current begins to fall exponentially to the normal steady state reverse saturation current. The time during which current falls to steady state reverse saturation is known as transition time as shown in figure above. The sum of the storage time and transition time is called reverse recovery time.

i.e. $t_{rr} = t_s + t_r$

Chapter: 2 (1 numerical + 1 theory) ⇒ 15 marks

Bipolar Junction Transistor :-

→ A transistor consists of two p-n junctions. It is formed by sandwiching either p-type or n-type semiconductor between a pair of opposite types. There are two types of transistor they are :-

- i) n-p-n transistor
- ii) p-n-p transistor.

When n-p-n transistor is composed of two n-type semiconductor separated by a thin section of p-type and p-n-p transistor is formed by two p-type semiconductor separated by thin section of n-type as shown in figure

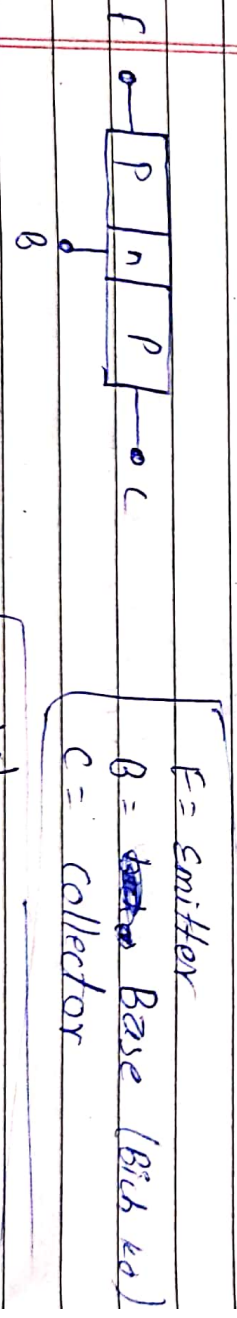
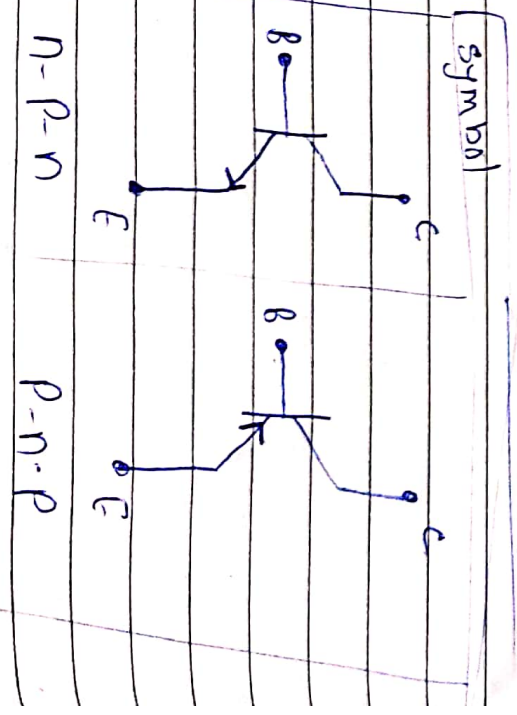
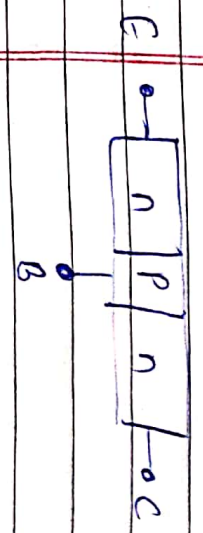


fig:- p-n-p transistor



E = emitter
B = Base (Bias)
C = collector

Transistor consist of three terminals mainly Emitter [E], Base [B], and collector [C]. The emitter region is highly doped, collector region is moderately doped. The emitter and base region is lightly doped. The emitter base junction is forward bias whereas the collector base junction is reverse bias.

* Working principle of n-p-n transistor

